One-loop contributions to the electroweak boson self-couplings as a tool to constrain BSM theories

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Abstract

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The CP-conserving and CP-violating ZZZ, $ZZ\gamma$, $Z\gamma\gamma$ and ZZW triple boson couplings were investigated at the one-loop level within both the SM and selected BSM models. In particular, the couplings h_3^Z, f_5^γ, and f_4^Z were computed within the MSSM and 2HDM frameworks for the proton-proton collisions with the center-of-mass energies up to 14 TeV, corresponding to the LHC range.

A general technique for constraining BSM theory parameters using experimental limits on anomalous couplings is proposed. This method is then applied to the 2HDM model, where parameter space constraints were derived using current LHC data.

1 Introduction

Nowadays, the main focus of experimental activities in the field of high energy physics is the comprehensive search for deviations from the Standard Model (SM). So-called Beyond the Standard Model (BSM) theories provide a basis for an extension of the SM towards some new, more general theory, explaining e.g. non-zero neutrino mass and existence of dark matter. Yet, the range of potential options for extending the Standard Model is wide. Experimental data play an essential role in highlighting feasible extensions of the Standard Model.

The LHC direct searches have found no evidence of new physics up to now, greatly narrowing the range of its possible manifestations [1–3]. Given the fact that the significant increase of collision energy at the LHC is not expected, the emergence of new results from direct searches is delayed for decades, when the future colliders appear.

On the contrary, the indirect searches have a great potential at the present time [4]. The idea of indirect search is that the existence of new physics at the inaccessible energy range has its low energy manifestations. There could be slight changes of the cross sections, branching ratios or coupling constants of the already known particles. Within the indirect searches, any deviation from the SM predictions can be treated in the context of model-independent approach, extending the SM Lagrangian with additional terms. The precision studies of the SM interactions using rare processes can take an advantage from the increase in luminosity, expected in the LHC Run3 and future HL-LHC [5]. Thus the use of current model-independent experimental limits becomes of critical importance for the probing of current BSM theoretical models.

The multiboson production is very sensitive tool to observe the manifestations of BSM physics effects. For example, the neutral gauge boson couplings such as $Z\gamma\gamma$, $ZZ\gamma$ and ZZZ are absent in the SM at tree level. However, BSM effects can lead to their appearance [6, 7]. Other couplings such as WWZ, besides SM part, would be also strengthened due to the BSM effects. The searches of such anomalous couplings have high priority for the leading high energy physics experiments such as ATLAS [8–10], CMS [11–13] and others. Besides the possible tree level BSM effects, one can expect mandatory loop-order contributions from heavy particles to these couplings. These loop-order effects are expected from both the SM and BSM theories.

The sensitivity of experimental results on anomalous couplings from the LHC can reach the accuracy of loop-order contributions from the SM or more likely some of BSM theories. In this case, the experimental limits on anomalous couplings can be translated into the direct limits on BSM theories' parameters and therefore constrain these theories.

Will be tuned after updating the paper structure Section 2 of the paper considers the one-loop effects in neutral couplings from SM and MSSM. An extension for the predictions for the SM and MSSM

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models in case of pp collisions is produced up to LHC energies ($\sqrt{s} = 14$ TeV). Section 3 of the paper presents a technique for the translation of the limits on anomalous couplings into the limits on BSM theory parameters. In section 4 the 2HDM theory parameters are constrained using LHC experiments limits on anomalous couplings based on developed technique. Section 5 contains discussion on the opportunities and limitations of such an approach.

2 Methodology

The technique for constraining the parameters of BSM theories used in this study is based on matching analytically computed one-loop amplitudes of particular processes with corresponding vertices of model-independent Lorentz decomposition, so-called vertex functions. In particular, within the neutral and charged triple-gauge vertices shown at Figure 1 the isolated CP-violating coefficient f_4^V ($V = Z, \gamma$) appearing in the standard tensor basis in ZZZ and ZWW interactions can be calculated and used together with the experimental limits on anomalous triple-gauge couplings to set the limits on model parameters.

For a chosen class of processes, the one-loop amplitude $\Gamma^{ZVV}_{\alpha\beta\mu}(P, q_1, q_2)$ can be computed and projected onto standard Lorentz structures that define the form factors $f_{4,5}^V$ within the vertex functions formalism [14, 15]. This projection is done via matching the $f_4^V(q^2; \vec{\Theta})$ coefficients of the characteristic Lorentz structures in computed theoretical one-loop amplitude and vertex function terms as functions of the momentum transfer q^2 and the model parameters $\vec{\Theta}$.

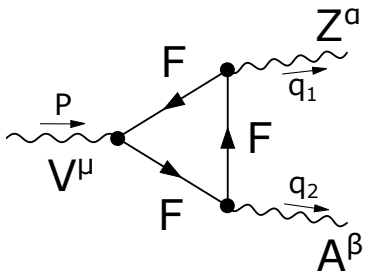


Figure 1: Fermionic triangle loop diagram.

The method includes the following steps:

1. **Parametrization.** Specifying a set of free model parameters $\vec{\Theta}$ and model-independent coefficients $\vec{f}$.
2. **Amplitude construction.** Generating all Feynman diagrams, contributing to the process of interest.
3. **Amplitude calculation.** Calculate the total analytical amplitude from all the Feynman diagrams contributing to the process.
4. **Matching.** Identifying the same Lorentz structures in both the theoretical amplitude and the model-independent decomposition and matching their coefficients to establish relations of the form $f_i = F_i(q^2; \vec{\Theta})$.
5. **Constraining.** Translating the experimental limits on the coefficients into constraints on the model parameters, i.e. $f_i = F_i(q^2; \vec{\Theta}) \leq f_i^{\text{exp}}$.

Since the theoretical prediction generally depends on the square of the momentum transfer, q^2, the results should either be obtained as functions of the momentum transfer or subjected to process-appropriate averaging, as discussed in Sec. 3.

In this study, for each vertex, the full set of one-loop diagrams with new BSM particles was generated via FeynArts [16]. The analytical amplitudes were built and simplified with FeynCalc [17]. Also, the experimental limits on model-independent coefficients f_4^V published in [12,18] were used. **Both one-dimensional slices and two-dimensional scans in (m_i, q) and (m_i, m_j) planes were provided. Where relevant, we display contours corresponding to current and projected integrated luminosities to indicate which parameter regions are excluded.**

3 Reduce dependence on q^2

In theoretical calculations, the vertex functions' form factors $f_4(q^2, \vec{\Theta})$ often depend on the squared momentum transfer q^2. This makes a direct comparison with experimental limits difficult, since experiments usually

provide constant upper limits. But such a comparison can still be performed by building a constant value from the general form factor that will be directly comparable to the experimental bound and then used to constrain model parameters.

In this study, three different methods to remove the explicit q^2-dependence from the $f_4(q^2, \vec{\Theta})$ form factor were probed:

1) **Fixing q^2 at a reference scale.** One can set q^2 at accelerator's reference scale ($q^2 = s_0$) and take $f_4^{\rm eff}(\vec{\Theta}) = f_4(s_0, \vec{\Theta})$. The choice of s_0 depends on the accelerator type. For e^+e^- colliders, $s_0 = s$. For hadron colliders, s_0 needs to be calculated (for example, as a spectrum-averaged value in the signal region).

2) **Cross-section matching.** In this method, the cross section is computed by using full q^2-dependent function $f_4(q^2, \vec{\Theta})$ with the same cuts and acceptance as in the experiment. Then the constant $f_4^{\rm eff}(\vec{\Theta})$ is obtained by requiring that the cross section computed with constant $f_4^{\rm eff}(\vec{\Theta})$ is equal to the first result **need to discuss this**. Within this method, q^2 is not fixed at the amplitude level.

3) **Spectrum-weighted averaging.** In this approach, the $f_4(q^2, \vec{\Theta})$ is averaged over q^2 with a weight that describe the shape of the event spectrum:

$$f_4^{\rm eff}(\vec{\Theta}) = \frac{\int w(q^2, \vec{\Theta})\, f_4(q^2, \vec{\Theta})\, dq^2}{\int w(q^2, \vec{\Theta})\, dq^2}, \qquad w(q^2) \propto \frac{d\sigma_{\rm theor}}{dq^2}. \tag{1}$$

The value $f_4^{\rm eff}(\vec{\Theta})$ is then compared directly to $f_4^{\rm exp}$. Below, all three methods are discussed in details. **Also need to discuss notations**

3.1 First method: fixing q^2 at a reference scale

The idea of this method is to fix the momentum transfer value inside the function $f_4(q^2)$ and thus get a constant evaluated at a reference scale s_0:

$$f_4^{\rm eff}(\vec{\Theta}) \equiv f_4(q^2 = s_0, \vec{\Theta}). \tag{2}$$

The reference scale depends on the accelerator and its kinematics and the process. Such approach removes the explicit q^2-dependence and allows a direct comparison to the experimental bound $f_4^{\rm exp}$.

For e^+e^- colliders, the reference scale is represented by the center-of-mass energy of the collisions ($s_0 = s$). For hadron collisions, the reference s_0 should be estimated from the additional analysis based on the probabilities of parton distribution functions (PDF). The same s_0 must be used later in the comparison to data.

After defining the scale, loop functions can be evaluated at $q^2 = s_0$. The resulting constant $f_4^{\rm eff}(\vec{\Theta})$ is then used in the setting the limits on model parameters $\vec{\Theta}$.

This method is simple and transparent and was used for example in [14]. If q^2 varies widely, then alternative procedures described below will provide more precise results.

3.2 Second method: averaging $f_4(q^2, \vec{\Theta})$ over q^2

This method replaces the form factor function $f_4(q^2)$ by a single constant obtained from a spectrum-weighted average. The weight follows the event yield, so that the most probable kinematics contribute the most to the average.

The weight is defined as the product of the partonic cross section and the parton luminosity,

$$w(q^2) \;\propto\; \hat{\sigma}(q^2)\, L(q^2). \tag{3}$$

The averaged value is then

$$\bar{f}_4(\vec{\Theta}) \equiv \frac{\displaystyle\int f_4(q^2, \vec{\Theta})\, \hat{\sigma}(q^2, \vec{\Theta})\, L(q^2)\, dq^2}{\displaystyle\int \hat{\sigma}(q^2, \vec{\Theta})\, L(q^2)\, dq^2} \;=\; \frac{\displaystyle\int_{\tau_{\rm min}}^1 \frac{1}{\tau}\, f_4(\tau s_0, \vec{\Theta})\, \hat{\sigma}(\tau s_0, \vec{\Theta})\, L(\tau)\, d\tau}{\displaystyle\int_{\tau_{\rm min}}^1 \frac{1}{\tau}\, \hat{\sigma}(\tau s_0, \vec{\Theta})\, L(\tau)\, d\tau}, \tag{4}$$

with $\tau = \hat{s}/s_0$ and $\tau_{\rm min} = 4M_Z^2/s_0$ for $q\bar{q} \to ZZ$. For e^+e^-, with fixed $q^2 = s_0$, the weight reduces to a delta function and $\bar{f}_4 = f_4(s_0, \vec{\Theta})$ (as in the first method).

The comparison with data is then done at the level of constants:

$$\bar{f}_4(\vec{\Theta}) = f_4^{\rm exp}. \tag{5}$$

The $\hat{\sigma}(\hat{s}, \vec{\Theta})$ **Why to use hats? These are not operators there is notation for parton cross section with hat (no pdf) and total without hat (with pdf)** and $L(\tau)$ were computed as in Sec. 3.3 and evaluated the two τ-integrals numerically (the same Monte Carlo setup can be used).

The results of this method depend on the chosen weight. Using the event yield is natural because it mirrors what the measurement is most sensitive to. **need to discuss model parameters**

3.3 Third method: comparison of total cross sections

The goal of this method is to compare the analytical cross section, either total or partial, with experimental limits that are calculated for a constant coupling. First, the theoretical cross section is computed by using form factor $f_4^{\text{eff}}(\vec{\Theta})$, calculated cross section will depend on model parameters $\vec{\Theta}$. Then, calculation of cross section with an equivalent constant f_4^{exp} was done; this cross section is model independent. Constrains on model parameters are determined by matching this cross sections:

$$\sigma\left(s_0; f_4^{\text{eff}}(\vec{\Theta})\right) \;\leq\; \sigma\left(s_0; f_4^{\text{exp}}\right) = \sigma_{\text{exp}}(s_0), \tag{6}$$

imposing the same cuts, acceptance, scales and PDFs on both sides. The constant $f_4^{\text{eff}}(\vec{\Theta})$ can then be compared directly to the experimental bound f_4^{exp}. This approach avoids any choice of a fixed q^2 at the amplitude level.

At hadron colliders (e.g. the LHC with $\sqrt{s_0}$), full cross section $\sigma(s_0)$ can be described (Eq. 7) through partonic cross section $\hat{\sigma}_{ab}(\hat{s})$ for $q\bar{q} \to ZZ$ (including the decays in case of applying fiducial cuts). $\hat{s}$ and $\hat{t}$ are partonic Mandelstam variables. μ_F^2 and μ_R^2 are factorization and renormalization scales and they are fixed to $\mu_F^2 \simeq \mu_R^2 \simeq \hat{s}$ partonic energy scale unless stated otherwise.

$$\sigma(s_0) \;=\; \sum_{ab} \int_{\tau_{\min}}^{1} \frac{1}{\tau}\, \hat{\sigma}_{ab}(\hat{s} = \tau s_0; \mu_F, \mu_R)\; L_{ab}(\tau; \mu_F) d\tau, \qquad \tau_{\min} = \frac{4M_Z^2}{s_0}, \tag{7}$$

where the parton luminosity is

$$L_{ab}(\tau; \mu_F) \;=\; \int_{\tau}^{1} \frac{1}{t}\, \text{xPDF}_a(t, \mu_F^2)\, \text{xPDF}_b\Big(\frac{\tau}{t}, \mu_F^2\Big)\, dt. \tag{8}$$

The partonic cross section is obtained from the squared matrix element, summed/averaged over spins and polarizations, and integrated over phase space:

$$\begin{aligned} t_{\text{lim}}^{\pm} &= -\frac{\hat{s} - 2M_Z^2}{2} \;\pm\; \frac{1}{2}\sqrt{\hat{s}^2 - 4\hat{s}M_Z^2}, \\ \hat{\sigma}_{ab}(\hat{s}) &= \frac{1}{16\pi\hat{s}^2} \int_{t_{\text{lim}}^-}^{t_{\text{lim}}^+} \big|\, M_{ab}(\hat{s}, \hat{t})\, \big|^2\, d\hat{t}, \end{aligned} \tag{9}$$

The scattering amplitudes $M_{ab}(\hat{s}, \hat{t})$ were computed with `FeynArts` and `FeynCalc`, including the projection that isolates the tensor structure of form factor f_4. Loop integrals were reduced to Passarino–Veltman functions. PDFs are evaluated by the `LHAPDF` with the `NNPDF3.0` set from PDF4LHC recommendations [19]. In the following, every hadronic cross section is understood to include the sum over all possible initial-state parton pairs, even when this sum is not written.

The phase-space and luminosity integrals are evaluated numerically using the built-in *AdaptiveQuasiMonteCarlo* algorithm in *Wolfram Mathematica*. For better sampling, a logarithmic grid was used. With $t = 10^y$ and $\tau = 10^k$ one gets

$$L_{ab}(\tau) = \ln 10 \int_{\log_{10}\tau}^{0} \text{xPDF}_a(10^y, \mu_F^2)\, \text{xPDF}_b\Big(\frac{\tau}{10^y}, \mu_F^2\Big)\, dy, \tag{10}$$

$$\sigma = \ln 10 \sum_{ab} \int_{\log_{10}\tau_{\min}}^{0} \hat{\sigma}_{ab}(10^k s_0; \mu_F, \mu_R)\; L_{ab}(10^k; \mu_F)\; dk. \tag{11}$$

However, there is a limitation of this approach connected to replacement theoretical coupling with its experimental value. The experimental amplitude has a different dependence on q^2: it does not include propagator terms, which normally introduce a suppression factor of the form $1/q^2$ in the amplitude. The lack of this factor results in rapid increase in the total cross section $\sigma_{\text{exp}}(s_0)$ with rising center-of-mass energy s_0. This behavior contrasts sharply with the theoretical formula, where the cross section decreases as s_0 increases. In the case of the experimental cross section, higher incoming energy leads to a larger cross section, whereas for the theoretical one, the opposite trend is observed. So the limits obtained by using this approach are very conservative.

4 Application to the 2HDM

The applicability of the proposed method is demonstrated through its application to the two-Higgs-doublet model (2HDM). This model adds a second scalar $SU(2)_L$ doublet. This leads to extra physical states and new bosonic interactions, which create loop-induced contributions to anomalous triple–gauge couplings. Thus, the loops contain both SM gauge bosons (Z_μ, $W^\pm_\mu$) and BSM scalars (G^0, $G^\pm$, h_i, $H^\pm$). General reviews can be found in [20–22]. The calculation is performed in the R_ξ gauge.

Within this model, the free parameters that contribute to ZZZ and ZWW vertices are: two mixing angles α_2, α_3 and three masses m_{h_2}, m_{h_3}, $m_{H^\pm}$. The remaining inputs are fixed, e.g. $v = 246$ GeV and $m_{h_1} = 125$ GeV. The parameters α_1 and $\tan\beta$ are also kept fixed (with $\beta = \alpha_1$ in this setup). The bosonic basis **What is bosonic basis?** is chosen to be consistent with Appendix A, and the conventions of [22] are used. Technical details (potential, mixing, and gauge interactions) are collected in the appendix.

The anomalous vertices are parametrized as

$$\Gamma^{\alpha\beta\mu}_{ZZV} = \frac{i(s - m_V^2)}{m_Z^2}\Big(f_4^V\left(P^\alpha g^{\mu\beta} + P^\beta g^{\mu\alpha}\right) - f_5^V \epsilon^{\mu\alpha\beta\rho}(q_1 - q_2)_\rho\Big), \tag{12}$$

for the neutral sector, and

$$\Gamma^{\alpha\beta\mu}_{ZWW} = i\Big(f_4^Z\left(P^\alpha g^{\mu\beta} + P^\beta g^{\mu\alpha}\right) - f_5^Z \epsilon^{\mu\alpha\beta\rho}(q_1 - q_2)_\rho\Big), \tag{13}$$

for the charged sector, with $P^\mu = q_1^\mu + q_2^\mu$ and $s = P^2$. The form factors f_4^V and f_5^V encode the loop effects. The CP–violating form factor f_4^Z, which contribute to ZZZ and ZWW vertices, is considered.

In the next subsections, the $f_4^Z(q^2, \vec{\Theta})$ form factors for ZZZ and ZWW vertices are calculated and the explicit q^2–dependence is removed using the three reduction methods from Sec. 3. As a result, a constant $f_4^{\text{eff}}(\vec{\Theta})$ value is obtained and compared to experimental limits to constrain the model parameters.

4.1 $Z\gamma\gamma$ and $ZZ\gamma$ vertices

In the 2HDM, one-loop diagrams with BSM scalars in the loop (neutral h_i, charged $H^\pm$ and Goldstone bosons $G^\pm, G^0$) *do not* generate the anomalous couplings h_3^V, h_4^V, f_4^γ or f_5^V ($V = Z, \gamma$) that are currently constrained by experiment.

The couplings $G^+G^-Z_\mu$, $G^+G^-A_\mu$, $H^+H^-Z_\mu$ and $H^+H^-A_\mu$ are proportional to $(q_+ - q_-)^\mu$, where $q_\pm$ are the four-momenta of the charged scalars. In triangle loops with three scalar lines, reversing the direction of the charge flow flips $(q_+ - q_-)^\mu \to -(q_+ - q_-)^\mu$. Summing both orientations therefore cancels the contribution. This cancellation holds for all three-scalar-loop topologies shown in Fig. 2.

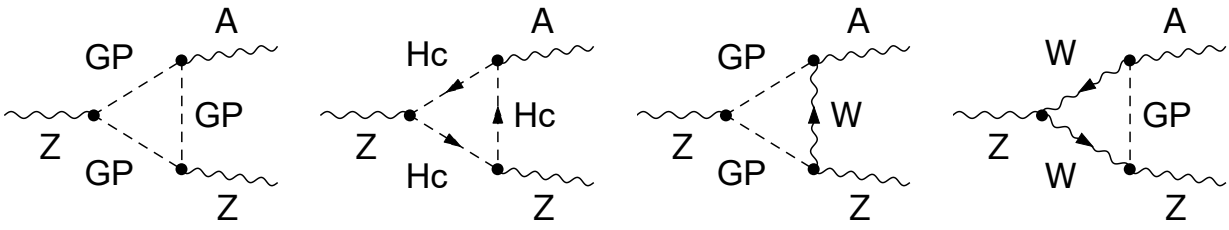


Figure 2: Representative one-loop topologies for $ZZ\gamma$ with 2HDM scalars in the loop.

The vertices $G^\pm W^\mp_\mu Z_\nu$ and $G^\pm W^\mp_\mu A_\nu$ are proportional to $g_{\mu\nu}$. Such terms cannot produce required anomalous tensor structures (e.g. antisymmetric $\epsilon_{\mu\alpha\beta\rho}(q_1 - q_2)^\rho$ or the $(P^\alpha g^{\mu\beta} + P^\beta g^{\mu\alpha})$ structures that define the couplings in Eqs. (12)–(13)). After summing over opposite charge-flow directions, these contributions also vanish.

For $Z\gamma\gamma$ and $ZZ\gamma$ with an on-shell photon, Ward identities enforce transversality ($q\cdot\varepsilon = 0$). Terms proportional to the total momentum P^μ or to external momenta contracted with the photon polarization drop out, further preventing the appearance of the required anomalous structures.

4.2 ZZZ vertex

The ZZZ vertex is sensitive to CP violation. In the 2HDM, the one-loop triangle diagrams that contribute to the CP-odd form factor f_4^Z contain neutral scalars h_i and the neutral Goldstone bosons G^0. All non-vanishing topologies are shown in Fig. 3.

The tree-level interactions that present in the loops are:

$$[h_i, h_j, Z^\mu] = \frac{g}{2c_W}\,(p_i - p_j)^\mu\,\epsilon_{ijk} X_{1k}, \tag{14}$$

$$[Z^\mu, G^0, h_i] = \frac{g}{2c_W}\,(p_i - p_0)^\mu\,X_{1i}, \tag{15}$$

$$[h_i, Z^\mu, Z^\nu] = i\,\frac{g}{c_W}\,m_Z\,g^{\mu\nu}\,X_{1i}, \tag{16}$$

where p_i is the incoming momenta and X_{1i} are the mixing factors (see Appendix A).

The three classes of loop amplitudes for $Z(q) \to Z(q_1)\,Z(q_2)$ can be written as

$$i\Gamma^{Z,hhh}_{\alpha\beta\mu} = -i\Big(\frac{g}{c_W}\Big)^3 T_{11}T_{12}T_{13}\int \epsilon_{ijk}\frac{k_\mu k_\alpha k_\beta}{\big((k-q_1)^2 - m_i^2\big)\big((k+q_2)^2 - m_j^2\big)\big(k^2 - m_k^2\big)}\frac{d^4k}{(2\pi)^4}, \tag{17}$$

$$i\Gamma^{Z,hhG}_{\alpha\beta\mu} = i\Gamma^{Z,hhh}_{\alpha\beta\mu}(m_i \leftrightarrow m_Z) + i\Gamma^{Z,hhh}_{\alpha\beta\mu}(m_j \leftrightarrow m_Z) + i\Gamma^{Z,hhh}_{\alpha\beta\mu}(m_k \leftrightarrow m_Z), \tag{18}$$

$$\begin{aligned} i\Gamma^{Z,hhZ}_{\alpha\beta\mu} = i\Big(\frac{g}{c_W}\Big)^3 T_{11}T_{12}T_{13}\int \epsilon_{ijk}\Bigg(&\frac{m_Z^2 k_\alpha g_{\mu\beta}}{\big((k-q_1)^2 - m_i^2\big)\big((k+q_2)^2 - m_Z^2\big)\big(k^2 - m_k^2\big)} - \\ &- \frac{m_Z^2 k_\beta g_{\mu\alpha}}{\big((k-q_1)^2 - m_Z^2\big)\big((k+q_2)^2 - m_j^2\big)\big(k^2 - m_k^2\big)}\Bigg)\frac{d^4k}{(2\pi)^4}. \end{aligned} \tag{19}$$

where R_ξ gauge was used with $m^2_{G^0} = m_Z^2$.

Projecting the full $\Gamma_{\alpha\beta\mu}$ on the Lorentz basis of Eq. (12) and reducing the tensors to scalar Passarino–Veltman (PV) functions gives the CP-odd form factor $f_4^Z(q^2)$:

$$\begin{aligned} e\,\frac{q^2 - m_Z^2}{m_Z^2}\,f_4^Z(q^2, m_i, \alpha_j) = &\left[\frac{1}{16\pi^2}\Big(\frac{g}{c_W}\Big)^3 T_{11}T_{12}T_{13}\right]\sum_{i,j,k}\epsilon_{ijk}\Big(-\,C_{001}(q^2, m_Z^2, m_Z^2; m_i^2, m_j^2, m_k^2) \\ &+\,C_{001}(q^2, m_Z^2, m_Z^2; m_i^2, m_j^2, m_Z^2) + C_{001}(q^2, m_Z^2, m_Z^2; m_Z^2, m_j^2, m_k^2) \\ &+\,C_{001}(q^2, m_Z^2, m_Z^2; m_i^2, m_Z^2, m_k^2) - \,m_Z^2\,C_1(q^2, m_Z^2, m_Z^2; m_i^2, m_Z^2, m_k^2)\Big), \end{aligned} \tag{20}$$

where C_{001} and C_1 are standard PV scalar functions. Ultraviolet poles cancel in the combination that defines f_4^Z, so only finite PV functions present in Eq. (20).

The dependence on the mixing angles arises through $T_{11}T_{12}T_{13} = \sin^2(\alpha_2)\cos(\alpha_2)\,\sin(\alpha_3)\cos(\alpha_3)$ and is independent of α_1 in the current setup. This factor rescales the overall size of $f_4^Z(q^2)$ but does not change its q^2 and mass dependencies.

The PV functions were calculated with `LoopTools` in *Mathematica*. The $f_4^Z(q^2)$ form factor was computed as a function of the parameters $(m_{h_2}, m_{h_3}, m_{H^\pm}, \alpha_2, \alpha_3)$. The momentum transfer q^2 is a kinematic variable and it was a subject to the q-reduction procedures described in Sec. 3. In this section, only the raw q^2-dependent result of Eq. (20) are shown.

4.3 ZWW vertex

The calculation of the CP-violating form factor $f_4^Z(q^2)$ for the ZWW vertex follows the same steps as for ZZZ vertex (projection and PV reduction; see Sec. 4.2 and Eqs. (12)–(13) for notation and tensor structure).

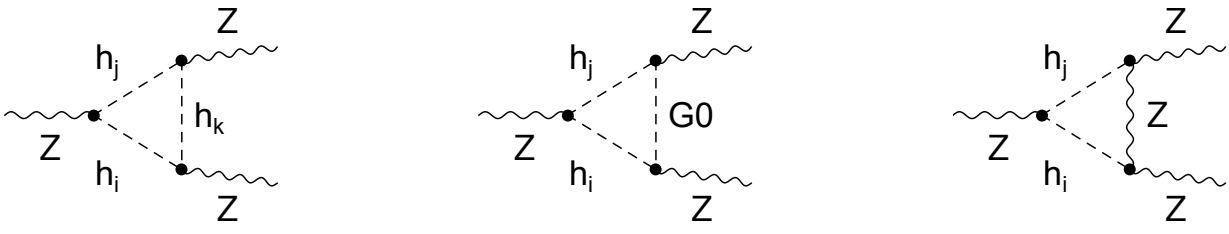


Figure 3: One-loop CP-violating ZZZ triangle topologies with 2HDM scalars in the loop.

The one-loop amplitude was then projected onto the CP-odd structure that defines f_4^Z in Eq. (13) and the result was reduced to scalar PV functions.

Compared to ZZZ, the ZWW vertex is important for three reasons: (i) it brings *explicit* sensitivity to the charged-Higgs mass $m_{H^\pm}$ through the loops, which ZZZ does not; (ii) **look branching for charged WW the WW final state has larger production rates and different kinematics this I don't understand**; *(iii) процесс в рамках СМ цп четный, но вклад в вершину цп-нечетный. Таким образом измерение цп в вершине также будет проявлением новой физики.* Hence ZWW and ZZZ together provide stronger and more robust constraints than either channel alone.

The tree-level interactions present in the loops are:

$$[h_i, G^+, W^{-\mu}] = i\frac{g}{2}\,(p_i - p_+)^\mu\, X_i, \tag{21}$$

$$[h_i, H^+, W^{-\mu}] = i\frac{g}{2}\,(p_i - p_+)^\mu\, Y_i, \tag{22}$$

$$[h_i, W^{+\mu}, W^{-\nu}] = i\,g\,M_W\,g_{\mu\nu}\,X_i, \tag{23}$$

$$[G^+, G^-, Z^\mu] = -\,i\frac{g}{2c_W}\cos 2\theta_W\,(p_+ - p_-)^\mu, \tag{24}$$

$$[H^+, H^-, Z^\mu] = -\,i\frac{g}{2c_W}\cos 2\theta_W\,(p_+ - p_-)^\mu. \tag{25}$$

All non-vanishing one-loop topologies built from these vertices are shown in Fig. 4.

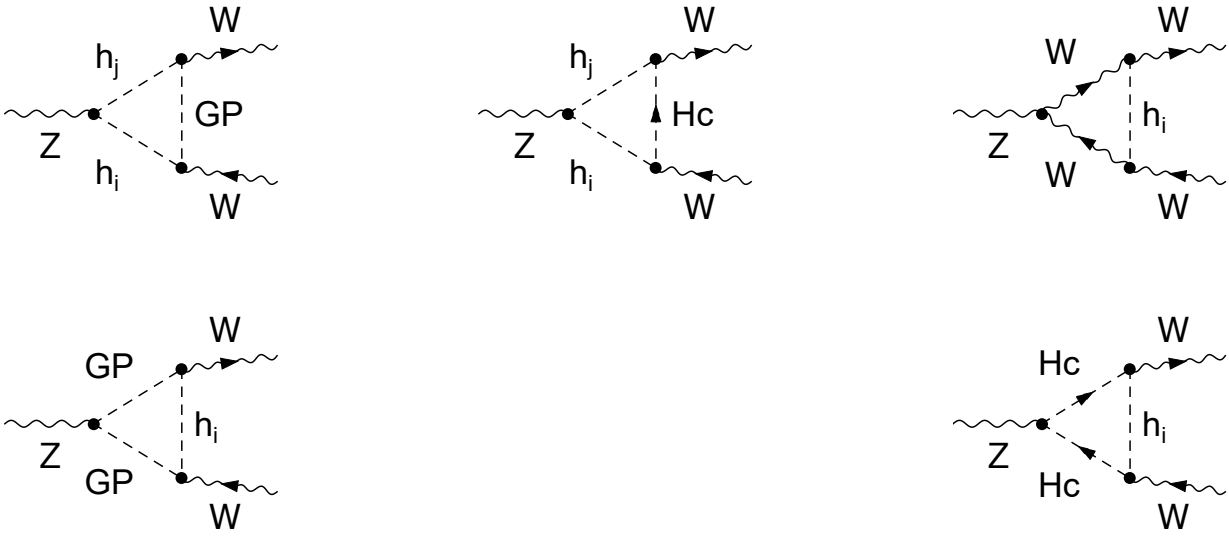


Figure 4: Representative one-loop ZWW triangle topologies in the 2HDM that contribute to the CP-odd form factor $f_4^Z(q^2)$.

After Passarino–Veltman reduction, the form factor can be written as (see also [23]):

$$f_4(q^2, m_i, \alpha_j) = \left(\frac{1}{16\pi^2 e}\Big(\frac{g}{c_W}\Big)^3 T_{11}T_{12}T_{13}\right)\sum_{i,j,k}\epsilon_{ijk}\Big[C_{001}(q^2, m_W^2, m_W^2, m_i^2, m_j^2, m_W^2) - C_{001}(q^2, m_W^2, m_W^2, m_i^2, m_j^2, m_{H^\pm}^2)\Big]. \tag{26}$$

with standard PV functions C_{001}. Ultraviolet poles cancel as in the ZZZ case, so Eq. (26) contains only finite combinations (cf. Sec. 4.2).

5 Results

This section converts the momentum-dependent one-loop predictions for the ZZZ and ZWW vertices into constant quantities that can be compared directly with experimental limits, and uses them to constrain the parameter spaces of the benchmark BSM scenarios introduced earlier. Unless stated otherwise, the setup described in Secs. 4.2 and 4.3 is used: collider energy $\sqrt{s_0} = 13.6$ TeV, the PDF set `NNPDF30_nnlo_as_0118`, and fixed model parameters:

$$m_{h_1} = 125\,\text{GeV};\ m_{h_3} = \sqrt{m_{h_2}^2 + v^2};\ \alpha_1 = 0.1;\ \alpha_2 = \arcsin\sqrt{2/3};\ \alpha_3 = \pi/4. \tag{27}$$

Since the theoretical predictions depend on q^2, the results are reported either as functions of the momentum transfer or in terms of process-specific constant reductions, as discussed in Sec. 3. All three methods for removing the explicit q^2 dependence are applied. One-dimensional slices and two-dimensional scans are presented in the (m_i, q) and (m_i, m_j) planes, where m_i denote the Higgs-boson masses in the 2HDM. Where applicable, contours corresponding to current and projected integrated luminosities are overlaid to indicate excluded regions of parameter space.

For the ZZZ vertex, published bounds on the coupling f_4^Z from Refs. [12,18] are quite weak. However the latest ATLAS $Z\gamma$ measurement results [24] on effective field theory parameters constraints can be transformed into the limits on f_4^Z using equations from [25]. Resulting limits were used (140 fb^{-1} $f_4^{\text{exp}} = 2.0 \times 10^{-4}$) and they were extrapolated towards future expected results: full LHC Run3 (300 fb^{-1} $f_4^{\text{exp}} = 1.3 \times 10^{-4}$) and HL-LHC Run (4000 fb^{-1} $f_4^{\text{exp}} = 3.7 \times 10^{-5}$). In the following, $f_{4,\text{ZZZ}}^{\text{exp}}$ denotes the experimental upper limit on $f_{4,\text{ZZZ}}$ (at a fixed confidence level and within the corresponding fiducial definition of the analysis), evaluated for the integrated luminosities considered in this work.

At present, no experimental limits on f_4^Z from measurements primarily sensitive to the ZWW vertex are available. The same reduction procedures can be applied once such bounds are obtained; for completeness, projected constraints based on assumed future limits in the ZWW channel are presented in Appendix B.

For the considered 2HDM parameterization, the dependence on the mixing angles enters the amplitude as an overall prefactor $T_{11}T_{12}T_{13}$, while the remaining kinematic and mass dependence is unchanged. The prefactor has a simple analytic maximum in magnitude,

$$\alpha_3 = \frac{\pi}{4}, \quad \alpha_2 = \arcsin\sqrt{\tfrac{2}{3}} \approx 0.955, \quad \left|T_{11}T_{12}T_{13}\right|_{\text{max}} = \frac{1}{3\sqrt{3}} \approx 0.192. \tag{28}$$

The overall-angle factor rescales the predicted f_4 values and therefore modifies the excluded regions at fixed f_4^{exp}. Defining $A \equiv T_{11}T_{12}T_{13}/\left|T_{11}T_{12}T_{13}\right|_{\text{max}}$, the form factor scales as $f_4 \to A\, f_4$ relative to the maximal-angle benchmark. Equivalently, at fixed prediction shape, the exclusion condition $|f_4| \le f_4^{\text{exp}}$ rescales as $f_4^{\text{exp}} \to f_4^{\text{exp}}/|A|$. An illustration of this rescaling is shown in Fig. 5. The results shown below use the mixing angles that maximize the prefactor to illustrate the best achievable sensitivity.

5.1 Method 1: Fixed momentum transfer

This method, described in Sec. 3.1, is most useful when the momentum transfer is fixed. This is the case, for example, at electron–positron colliders, where the squared momentum transfer equals the collider center-of-mass energy, $q^2 = s_0$. One can then directly compare the resulting constant value of f_4 with experimental bounds obtained at the same energy **change places for ZZZ and exp, ZZZ up exp down**:

$$\left|f_{4,\text{ZZZ}}(q^2 = s_0, \vec{\Theta})\right| \le f_{4,\text{ZZZ}}^{\text{exp}}\,. \tag{29}$$

The allowed region in the parameter space $\vec{\Theta}$ is defined by this inequality, while parameter points with $\left|f_{4,\text{ZZZ}}(s_0, \vec{\Theta})\right| > f_{4,\text{ZZZ}}^{\text{exp}}$ are excluded.

The application of this method to the neutral ZZZ vertex is illustrated in Figs. 6 and 8. Figure 6 shows $\left|f_{4,\text{ZZZ}}\right|$ as a function of the momentum transfer q for several values of m_{h_2}, with fixed mixing angles and with the benchmark relation $m_{h_3} = \sqrt{m_{h_2}^2 + v^2}$ imposed. The horizontal lines represent the experimental bounds $f_{4,\text{ZZZ}}^{\text{exp}}$ for the considered luminosity scenarios.

The complementary two-dimensional map in Fig. 8 presents the same constraint in the (m_{h_2}, m_{h_3}) plane at $q^2 = s_0$. The contour defined by

$$\left|f_{4,\text{ZZZ}}(q^2 = s_0, m_{h_2}, m_{h_3})\right| = f_{4,\text{ZZZ}}^{\text{exp}} \tag{30}$$

marks the boundary between the allowed and excluded regions; points inside this contour in $\left|f_{4,\text{ZZZ}}\right|$ are excluded.

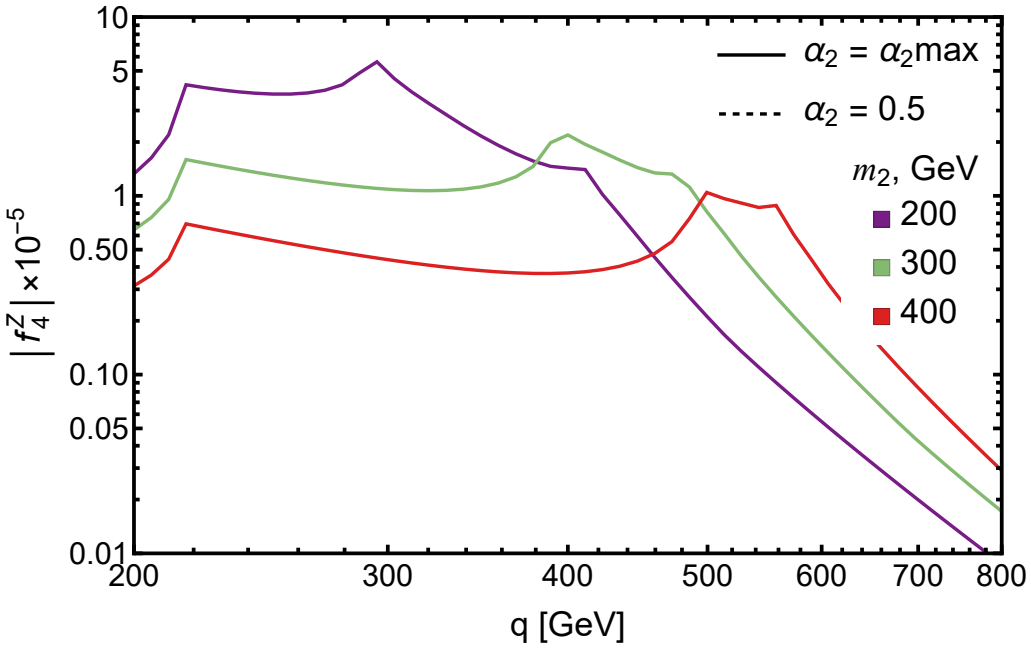


Figure 5: Rescaling of $|f_{4,\text{ZZZ}}|$ with mixing angles for a benchmark mass point. Solid lines correspond to α_2 that maximize amplitude and the dashed lines correspond to $\alpha_2 = 0.5$.**The style of the plot is still 1 y.o. Should be changed completely. And pretty much all the plots below убрать налезания 200 и 0.01 set m2 legend left down corner**

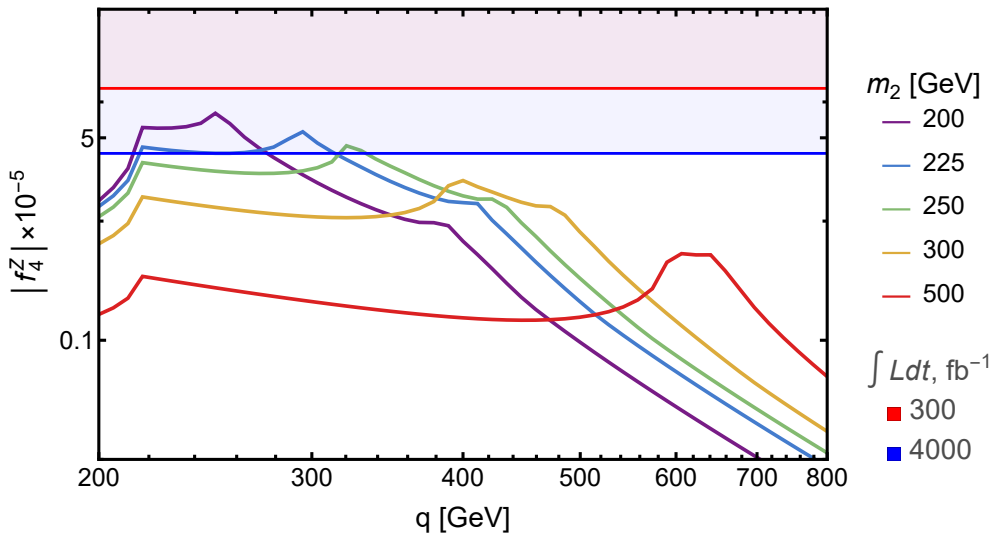


Figure 6: $|f_{4,\text{ZZZ}}|$ as a function of q for different m_{h_2} (with all other model and setup parameters fixed as specified in Eq. 27.). **Сделать горизонтальные легенды внутри графиков. Вернуть тики** The horizontal line marks the experimental limit $f_{4,\text{ZZZ}}^{\text{exp}}$ at $\sqrt{s_0}$.

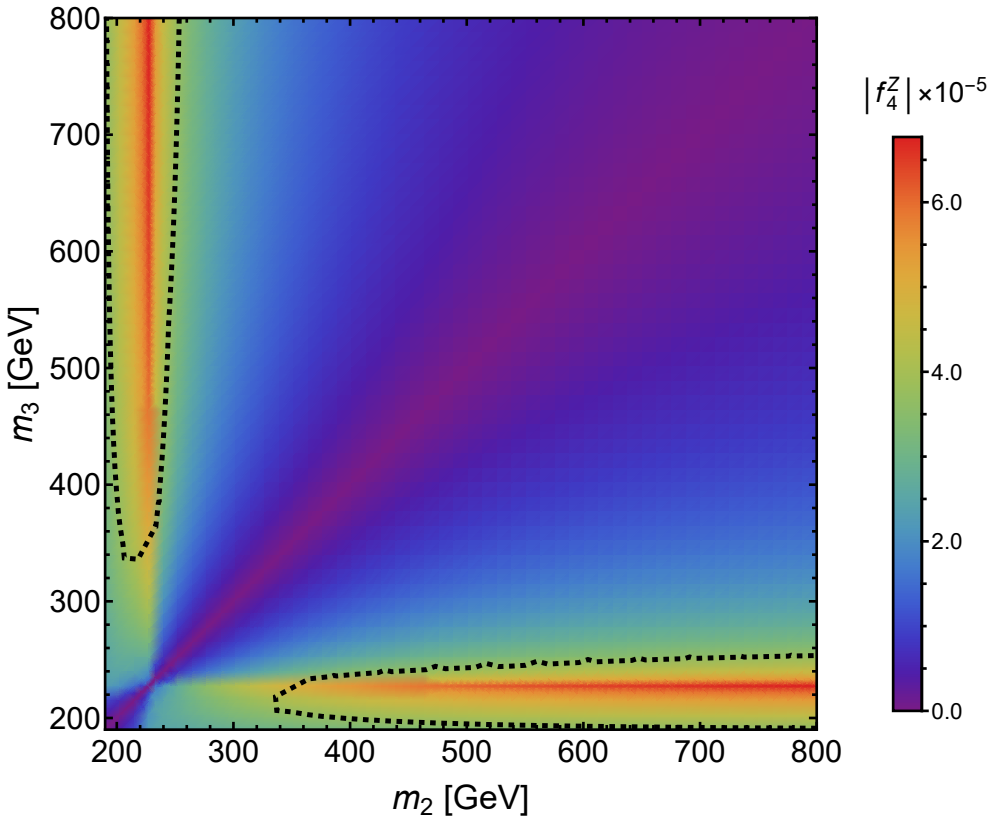


Figure 7: Exclusion map of $|f_{4,\mathrm{ZZZ}}|$ in the (m_{h_2}, m_{h_3}) plane at $q = 320\,\mathrm{GeV}$. All other model and setup parameters fixed as specified in Eq. 27. **Легенду в полный рост, в два раза шире. Шрифты легенды сделать равными. Надпись справа по центру** The dashed contour $|f_{4,\mathrm{ZZZ}}| = f_{4,\mathrm{ZZZ}}^{\mathrm{exp}}$ is indicated.

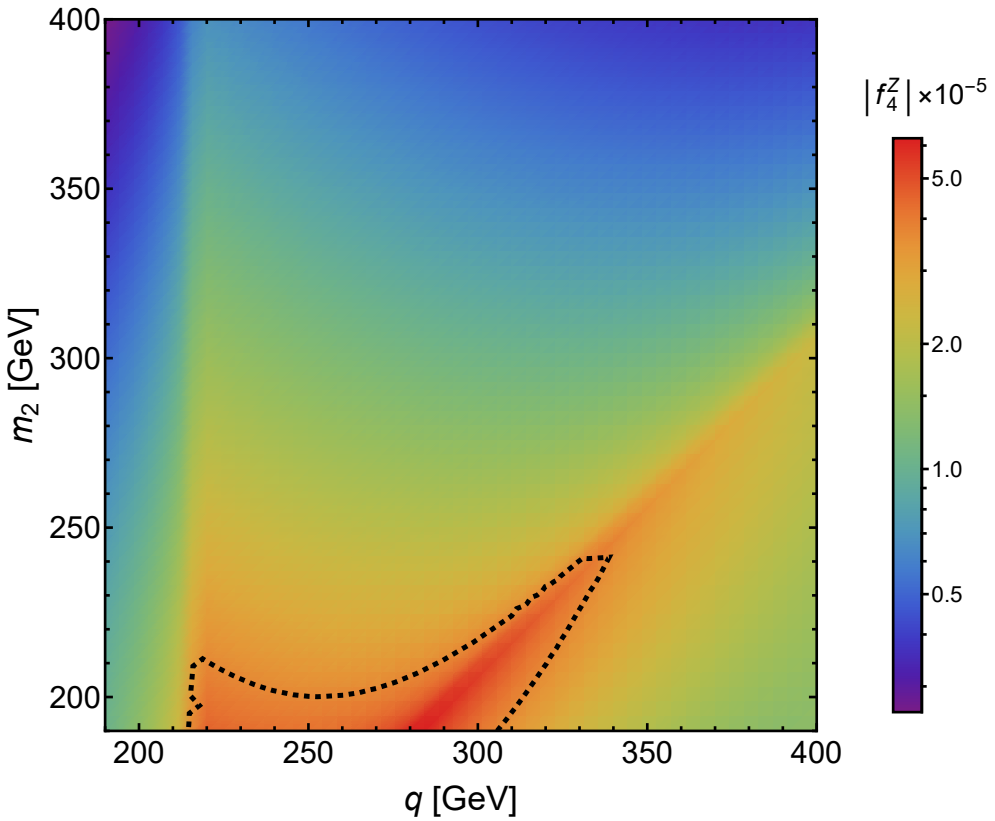


Figure 8: Exclusion map of $|f_{4,\mathrm{ZZZ}}|$ in the (q, m_{h_2}) plane with all other model and setup parameters fixed as specified in Eq. 27. The dashed contour $|f_{4,\mathrm{ZZZ}}| = f_{4,\mathrm{ZZZ}}^{\mathrm{exp}}$ is indicated.

Table 1: Constraints on the mass parameter m_{h_2} derived from the intersection roots at different values of q for an luminosity of $4000\,\mathrm{fb}^{-1}$. All other model and setup parameters fixed as specified in Eq. 27.

q [GeV]	m_{h_2} [GeV] at $\mathcal{L} = 4000\,\mathrm{fb}^{-1}$
300	Unconstrained
350	Unconstrained
400	$198.9 \leq m_{h_2} \leq 246.5$
450	$m_{h_2} \leq 303.3$
500	$200.1 \leq m_{h_2} \leq 351.6$
550	$235.2 \leq m_{h_2} \leq 396.7$
600	$285.8 \leq m_{h_2} \leq 439.5$
650	$365.8 \leq m_{h_2} \leq 479.9$
700	Unconstrained

The constraints on the model parameter space can be extracted directly from these figures and some values are shown at Table 1.

This fixed-scale approach is simple and transparent. When the relevant q^2 spectrum is broad, as in typical hadron-collider production, methods based on q^2 averaging or observable matching provide a more process-specific reduction of the q^2 dependence.

5.2 Method 2: Spectrum-weighted average

In this subsection, a constant effective value is obtained by averaging the momentum-dependent form factor $f_{4,\mathrm{ZZZ}}(q^2, \vec{\Theta})$ over the relevant kinematic phase space. The average is taken over the partonic invariant mass $q^2 = \hat{s} = \tau s_0$, with $\tau \in [\tau_{\min}, 1]$. A detailed description and derivation of the averaging procedure are given in Sec. 3.2.

In practice, the weighting can be based on the cross section of the process $\sigma(q^2, \vec{\Theta})$, and the luminosity $L(q^2)$ or detector effects such as acceptance $\mathrm{A}(q^2)$. In this analysis, detector acceptance is not applied. The phase space is defined by the chosen kinematic integration and PDFs. In other applications, experimental fiducial cuts can be implemented by restricting the integration region and, if needed, by adding an acceptance factor $A(q^2)$ to the weight.

In this work, the weight is chosen proportional to the product of the partonic cross section and the parton luminosity:

$$\bar{f}_{4,\mathrm{ZZZ}}(\vec{\Theta}) \equiv \frac{\sum_{a,b} \int_{\tau_{\min}}^{1} \frac{d\tau}{\tau} f_{4,\mathrm{ZZZ}}(\tau s_0, \vec{\Theta})\, \hat{\sigma}_{ab}(\tau s_0)\, L_{ab}(\tau)}{\sum_{a,b} \int_{\tau_{\min}}^{1} \frac{d\tau}{\tau} \hat{\sigma}_{ab}(\tau s_0)\, L_{ab}(\tau)}\,, \tag{31}$$

where L_{ab} is the parton luminosity and $\hat{\sigma}_{ab}$ is the partonic cross section (see Eqs. 11 and 10). The indices a, b label the initial-state quark flavours in the $q\bar{q} \to ZZ$ channels kept in the computation. In the numerical implementation, only the dominant $u\bar{u}$ and $d\bar{d}$ channels are included. By construction, $\bar{f}_{4,\mathrm{ZZZ}}(\vec{\Theta})$ depends only on the model parameters $\vec{\Theta}$ and the analysis setup that defines s_0, $\hat{\sigma}_{ab}$, and $L_{ab}(\tau)$.

The averaged coupling $\bar{f}_{4,\mathrm{ZZZ}}(\vec{\Theta})$ can then be used to constrain the model parameters $\vec{\Theta}$ through

$$\left|\bar{f}_{4,\mathrm{ZZZ}}(\vec{\Theta})\right| \;\leq\; f_{4,\mathrm{ZZZ}}^{\mathrm{exp}}\,. \tag{32}$$

For a fixed momentum transfer, the weight reduces to a delta function in q^2, and the present method reduces to Method 1.

Figure 14 shows $\left|\bar{f}_{4,\mathrm{ZZZ}}(\vec{\Theta})\right|$ as a function of m_{h_2} at fixed $\sqrt{s_0}$. The horizontal line indicates the experimental bound $f_{4,\mathrm{ZZZ}}^{\mathrm{exp}}$. Intersection points with this line, if present, determine the excluded and allowed ranges of m_{h_2} for the chosen benchmark inputs.

Figure 15 compares $\left|\bar{f}_{4,\mathrm{ZZZ}}(\vec{\Theta})\right|$ as a function of m_{h_2} for several values of $\sqrt{s_0}$. This comparison illustrates the stability of the qualitative m_{h_2} dependence against variations of the collider energy in the tested range; the overall normalization changes through both $\hat{\sigma}_{ab}$ and $L_{ab}(\tau)$.

Within Method 2, excluding a visible part of the parameter space would require improving the experimental sensitivity to the $\mathcal{O}(10^6)$ level for f_4^{exp} (order-of-magnitude estimate). The exact threshold depends on $\sqrt{s_0}$ and weighting details in Eq. (31).**Need to add few words about other possible cases**

5.3 Method 3: Cross-section matching

In this approach, described in Sec. 3.3, the cross section is computed using the full q^2-dependent one-loop form factor and is compared to a cross-section limit derived from a constant-coupling bound. For each benchmark point, the cross section $\sigma_{\text{full}}(s_0; f_4^Z(q^2, \vec{\Theta}))$ is evaluated in the same setup as in Sec. 3.3. The reference quantity $\sigma_{\text{exp}}(s_0)$ is not a measured cross section. Instead, it is obtained by using the same cross-section definition, cuts, scales, and PDFs, but replacing the q^2-dependent form factor by a constant coupling set to the experimental bound, $f_4^Z \to f_4^{\text{exp}}$. This translation provides an upper limit on the cross section that corresponds to the published constraint on f_4^{exp}.

Constraints on the model parameters $\vec{\Theta}$ are then defined through the matching condition in Eq. (6),

$$\sigma_{\text{full}}(s_0; f_4^Z(q^2, \vec{\Theta})) \;\leq\; \sigma_{\text{exp}}(s_0)\,, \tag{33}$$

with identical analysis choices on both sides.

The outcome of this procedure is shown in Figs. 9 and 10. In the benchmark scans considered here, the predicted cross sections remain below the corresponding upper limits $\sigma_{\text{exp}}(s_0)$ (lines derived from f_4^{exp}), including the projected sensitivity scenarios used for the overlays. As a result, Eq. (33) does not produce exclusions in the explored parameter ranges, and no constraints are obtained from Method 3 for these scans. For the ZWW channel, the reference bounds used to build $\sigma_{\text{exp}}(s_0)$ follow the assumptions stated in Appendix B.

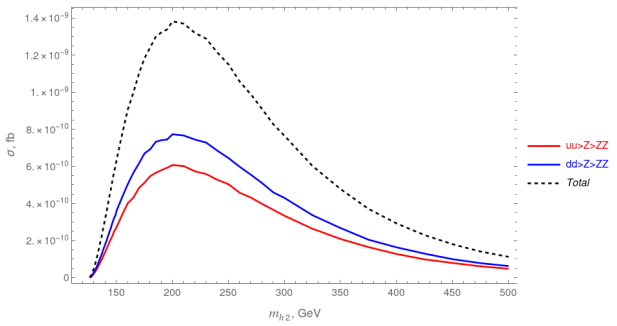


Figure 9: Cross section $\sigma(pp \to ZZ)$ as a function of the mass m_{h_2}, with all other model and setup parameters fixed as specified in Eq. 27. Colored curves show the contributions from the dominant initial-state parton channels; the black curve is their sum, i.e. the total cross section in the approximation used. Horizontal lines indicate the upper limits $\sigma_{\text{exp}}(s_0)$ derived from the current and projected bounds on f_4^{exp} at $\sqrt{s_0}$. **And where is the experimental limit?**

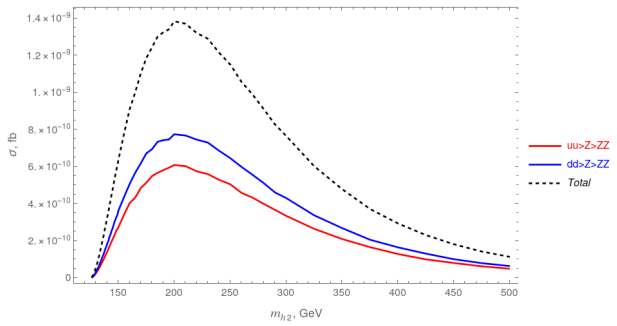


Figure 10: Same as in Fig. 9, but for the channel sensitive to the ZWW vertex. The total cross section stays below the corresponding upper limits $\sigma_{\text{exp}}(s_0)$ (derived from the current and projected bounds on f_4^{exp}) throughout the scan, and no crossing is observed.

6 Conclusion / Discussion

The predictions proposed in the paper [26] were extended up to 14 TeV in order to compare them with the existing LHC constraints. The proposed approach allows us to set the constraints on the free parameters of the BSM models. Predictions made in this paper are directly applicable to the searches for anomalous couplings, which are being carried out by the LHC experiments. In case of SM and MSSM the expected values for coupling parameters on the order of $10^{-8} \div 10^{-7}$ will be likely accessible on HL-LHC. Approach developed here is also applicable for other different models. Thus the considered approach shows great potential and can be used to constrains other BSM theories.**This we also need to discuss**

Acknowledgements

The authors also would like to thank N.L. Belyaev, S.R. Slabospitsky for fruitful discussions.

The work was funded by the Ministry of Science and Higher Education of the Russian Federation, Project "Studying physical phenomena in the micro- and macro-world to develop future technologies" FSWU-2026-0010.

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$$\mathrm{pp} \to \mathrm{ZZ}$$

production cross sections and constraints on anomalous triple gauge couplings at

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A 2HDM parametrization

According to [22], the scalar potential within the 2HDM model can be written as follows:

$$V(\Phi_1,\Phi_2) = Y_{a\bar{b}}\Phi_{\bar{a}}^{\dagger}\Phi_b + \frac{1}{2}Z_{a\bar{b}c\bar{d}}(\Phi_{\bar{a}}^{\dagger}\Phi_b)(\Phi_{\bar{c}}^{\dagger}\Phi_d), \quad j=1,2. \tag{34}$$

Here Φ_1 and Φ_2 are Higgs doublets; $a,b,c,d = \bar{a},\bar{b},\bar{c},\bar{d} = 1,2$; $Z_{a\bar{b}c\bar{d}}$ are, in general, complex model parameters and the summation is performed over barred with unbarred indices. The Higgs doublets themselves can be parameterized as:

$$\Phi_j = e^{i\xi_j}\begin{pmatrix} \varphi_j^{+} \\ (v_j+\eta_j+i\chi_j)/\sqrt{2} \end{pmatrix}, \quad j=1,2, \tag{35}$$

where v_i are the vacuum expectation values ($v_1^2+v_2^2=v^2$), η_i and φ_i are real fields and ξ_j are the Higgs phases.

Considering these notations, it is possible to define the following orthogonal states:

$$\begin{pmatrix} G_0 \\ \eta_3 \end{pmatrix} = \begin{pmatrix} v_1/v & v_2/v \\ -v_2/v & v_1/v \end{pmatrix}\begin{pmatrix} \chi_1 \\ \chi_2 \end{pmatrix} \tag{36}$$

and

$$\begin{pmatrix} G^{\pm} \\ H^{\pm} \end{pmatrix} = \begin{pmatrix} v_1/v & v_2/v \\ -v_2/v & v_1/v \end{pmatrix}\begin{pmatrix} \varphi_1^{\pm} \\ \varphi_2^{\pm} \end{pmatrix}. \tag{37}$$

The G_0 and $G^{\pm}$ correspond to Goldstone bosons and $H^{\pm}$ correspond to charged Higgs boson. The model also consists three scalar states which are linear combinations of η_i:

$$\begin{pmatrix} h_1 \\ h_2 \\ h_3 \end{pmatrix} = R\begin{pmatrix} \eta_1 \\ \eta_2 \\ \eta_3 \end{pmatrix}, \tag{38}$$

where the mixing matrix R is defined as:

$$R = \begin{pmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{pmatrix} = \begin{pmatrix} c_1\,c_2 & s_1\,c_2 & s_2 \\ -(c_1\,s_2\,s_3+s_1\,c_3) & c_1\,c_3-s_1\,s_2\,s_3 & c_2\,s_3 \\ -c_1\,s_2\,c_3+s_1\,s_3 & -(c_1\,s_3+s_1\,s_2\,c_3) & c_2\,c_3 \end{pmatrix}\cdot\begin{pmatrix} c_\beta & -s_\beta & 0 \\ s_\beta & c_\beta & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{39}$$

Thus mixing matrix defines the relation between new scalar states and, thus, the scale of CP-violation. Here $s_i = \sin\alpha_i, c_i = \cos\alpha_i$ and without loss generality $-\pi/2 \le \alpha_i \le \pi/2$. The values α_i are the mixing angles of new Higgs boson states.

$$T = \begin{pmatrix} c_1 & -s_1 & 0 \\ s_1 & c_1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} c_2 & 0 & -s_2 \\ 0 & 1 & 0 \\ s_2 & 0 & c_2 \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 \\ 0 & c_3 & -s_3 \\ 0 & s_3 & c_3 \end{pmatrix} \tag{40}$$

B ZWW analysis with assumed limits

Current LHC results provide direct bounds on several anomalous triple gauge couplings in channels with ZWW production. However, public limits are typically reported in a basis of effective couplings [13, 27] that differs from the form-factor notation used in this work. In particular, no published bound is currently available for the specific CP-violating form factor defined here as $f_{4,\mathrm{ZWW}}(q^2,\vec{\Theta})$ obtained from the one-loop ZWW vertex projection. As a result, the ZWW study in this appendix relies on specific assumption and is purely illustrative.

The ZWW vertex is of additional interest because, in the considered 2HDM benchmarks, the corresponding one-loop form factor depends not only on the neutral-scalar mass m_{h_2} but also on the charged-scalar mass $m_{H^{\pm}}$. As a result, any constraint derived from the neutral-sector scans in m_{h_2} can be propagated to the charged sector and translated into an allowed range for $m_{H^{\pm}}$ once the functional dependence $f_{4,\mathrm{ZWW}}(q^2; m_{h_2}, m_{H^{\pm}}, \ldots)$ is specified.

To define a benchmark sensitivity, the assumed bound on the constant coupling is fixed to a reference value denoted by $f_{4,\mathrm{ZWW}}^{\mathrm{exp}}$. In this work, the same numerical limit values as used for the ZZZ overlays at the corresponding luminosity scenarios are adopted,

$$f_{4,\mathrm{ZWW}}^{\mathrm{exp}} \equiv f_{4,\mathrm{ZZZ}}^{\mathrm{exp}}, \tag{41}$$

and are treated as inputs in the comparisons below. The impact of other choices is a simple rescaling of the exclusion contours.

B.1 Method 1: Fixed transferred momentum

Method 1 evaluates the momentum-dependent form factor at a fixed reference scale and compares it to a constant bound. The exclusion condition is

$$\left|f_{4,\mathrm{ZWW}}(q^2 = s_0, \vec{\Theta})\right| \leq f^{\mathrm{exp}}_{4,\mathrm{ZWW}}\,, \tag{42}$$

so that parameter points with $\left|f_{4,\mathrm{ZWW}}(s_0, \vec{\Theta})\right| > f^{\mathrm{exp}}_{4,\mathrm{ZWW}}$ are excluded.

Figure 11 shows $\left|f_{4,\mathrm{ZWW}}\right|$ as a function of the momentum transfer q for several values of m_{h_2} at fixed $m_{H^\pm}$. Figure 12 shows the dependence on $m_{H^\pm}$ at fixed m_{h_2} for a representative fixed value of q. In both panels, horizontal lines correspond to the assumed bounds $f^{\mathrm{exp}}_{4,\mathrm{ZWW}}$ for the considered luminosity scenarios. Intersection points, if present, determine the allowed ranges of the scanned mass parameter for that scenario.

The two-dimensional map in Fig. 13 summarizes the same constraint in the $(m_{h_2}, m_{H^\pm})$ plane at the fixed reference scale. The contour defined by $\left|f_{4,\mathrm{ZWW}}\right| = f^{\mathrm{exp}}_{4,\mathrm{ZWW}}$ marks the boundary between the allowed and excluded regions.

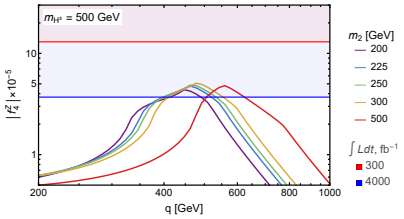


Figure 11: $|f^{\mathrm{eff}}_{4,\mathrm{ZWW}}|$ as a function of q for different m_{h_2} at fixed $m_{H^\pm}$.

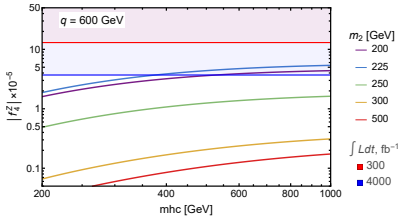


Figure 12: $|f^{\mathrm{eff}}_{4,\mathrm{ZWW}}|$ as a function of $m_{H^\pm}$ at fixed m_{h_2}.

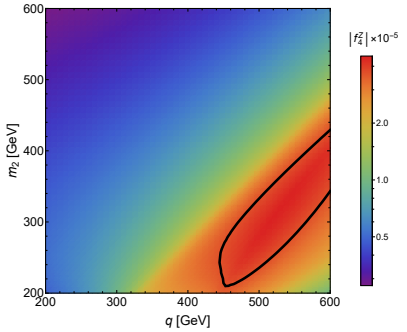


Figure 13: Exclusion map of $|f^{\mathrm{eff}}_{4,\mathrm{ZWW}}|$ in the $(\sqrt{s_0}, m_{h_2})$ plane at fixed $m_{H^\pm}$. The contour $|f^{\mathrm{eff}}_{4,\mathrm{ZWW}}| = f^{\mathrm{exp}}_{4,\mathrm{ZWW}}$ is shown.

B.2 Method 2: Spectrum-weighted average

Method 2 replaces the full momentum dependence by a constant effective value obtained from a spectrum-weighted average over $q^2 = \hat{s} = \tau s_0$, as described in Sec. 3.2. The weight is chosen proportional to the expected event yield in the selected setup, i.e. to the product of the partonic cross section and parton luminosity. The same numerical implementation is used as in the main text; only the dominant initial-state channels included in the code are retained.

The averaged coupling $\bar{f}_{4,\mathrm{ZWW}}(\vec{\Theta})$ is then compared to the assumed bound through

$$\left|\bar{f}_{4,\mathrm{ZWW}}(\vec{\Theta})\right| \leq f^{\mathrm{exp}}_{4,\mathrm{ZWW}}\,. \tag{43}$$

The corresponding one-dimensional scans and their comparison to the horizontal bound line follow the same reading as in Sec. 5.2.

Figure 14 shows $\left|\bar{f}_{4,\mathrm{ZWW}}(\vec{\Theta})\right|$ as a function of m_{h_2} at fixed $\sqrt{s_0}$. The horizontal line indicates the experimental bound $f^{\mathrm{exp}}_{4,\mathrm{ZWW}}$. Intersection points with this line, if present, determine the excluded and allowed ranges of m_{h_2} for the chosen benchmark inputs.

Figure 15 compares $\left|\bar{f}_{4,\mathrm{ZWW}}(\vec{\Theta})\right|$ as a function of m_{h_2} for several values of $\sqrt{s_0}$. This comparison illustrates the stability of the qualitative m_{h_2} dependence against variations of the collider energy in the tested range; the overall normalization changes through both $\hat{\sigma}_{ab}$ and $L_{ab}(\tau)$.

Figure 16 scans $m_{H^\pm}$ at fixed $\sqrt{s_0}$. The spectrum-weighted average varies only mildly over $m_{H^\pm} \in [400,\ 800]$,GeV for the chosen setup, and the ordering of the curves indicates that the charged-scalar mass affects $\bar{f}_4$ primarily through loop thresholds rather than through sharp kinematic enhancements. As in the previous cases, no crossing of the experimental line is observed.

B.3 Method 3: Cross-section matching

Method 3 compares the cross section computed with the full q^2-dependent one-loop form factor to an upper-limit cross section derived from the assumed constant-coupling bound. The reference quantity $\sigma_{\mathrm{exp}}(s_0)$ is

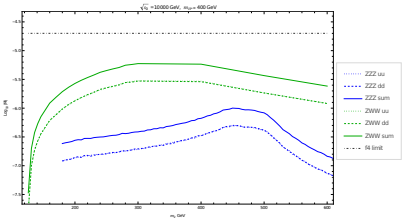


Figure 14: Spectrum-weighted average $\bar{f}_4$ as a function of m_{h_2} at fixed $\sqrt{s_0}$ and $m_{H^\pm}$. Curves show ZZZ, ZWW, and their sum. The horizontal line marks $f_4^{\rm exp}$.

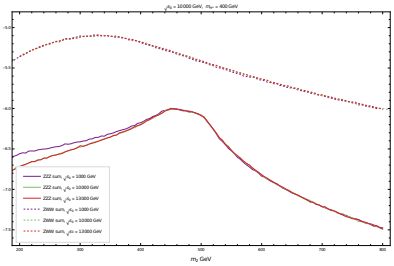


Figure 15: Dependence of $\bar{f}_4$ on $\sqrt{s_0}$ for fixed $m_{H^\pm}$. For each $\sqrt{s_0}$, $\bar{f}_4$ is shown vs. m_{h_2}; ZZZ and ZWW sums are indicated. The horizontal line marks $f_4^{\rm exp}$.

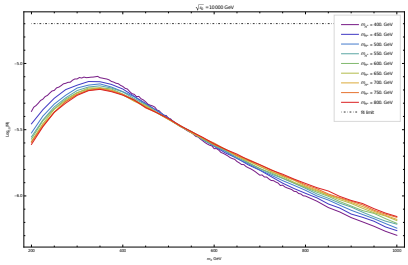


Figure 16: Charged-Higgs scan: $\bar{f}_4$ as a function of m_{h_2} for several $m_{H^\pm}$ values at fixed $\sqrt{s_0}$. The horizontal line marks $f_4^{\rm exp}$.**All the captions we need to revise aster finalization of the plots.**

obtained by computing the same cross section with the replacement $f_{4,\rm ZWW}(q^2,\vec{\Theta}) \to f_{4,\rm ZWW}^{\rm exp}$, using the same cuts, scales, and PDFs. The constraint condition is

$$\sigma_{\rm full}\Big(s_0; f_{4,\rm ZWW}(q^2,\vec{\Theta})\Big) \le \sigma_{\rm exp}(s_0)\,. \tag{44}$$

For the benchmark scans considered here, the predicted cross sections remain below the corresponding $\sigma_{\rm exp}(s_0)$ lines built from the assumed limits, and no exclusions are obtained in the explored parameter ranges.

C Application to the MSSM

In this section the possibility to constrain the MSSM parameters by using observational data [8,12], obtained by the LHC experiments, is investigated.

The variation of M_{χ_1} and M_{χ_2} in equation 48 affects the Charginos' contribution to the h_3^γ, h_3^Z and f_5^γ couplings. The goal is to study the possibility to constrain the characteristic energy scale parameter M from the experimental limits on the couplings.

The absolute values of the couplings h_3^γ, h_3^Z and f_5^γ as a function of energy scale parameter M are shown in Figure 18. These results show that one cannot constrain MSSM parameters using limits on anomalous couplings, since the latter are not strong enough yet.

The results for SM and MSSM are obtained using approach developed in the paper [26]. **we need to provide some description here** The previous result was reproduced and extended up to LHC collision energies. The values for CP-conserving neutral couplings h_3^γ, h_3^Z and f_5^γ in the $\gamma \to Z\gamma$, $Z \to Z\gamma$ and $\gamma \to ZZ$ decays from the SM and MSSM one-loop contributions (Figure **??**) are estimated as a function of $\sqrt{s}$. These contributions can become noticeable in the experiment since the experimental sensitivity increasing with the growth of collected integrated luminosity.

These neutral couplings can be probed at the LHC experiments and constrained, so that the expected values at high collision energies become important especially if the size of the loop contributions to the couplings is close to the existing limits.

The effective couplings h_3^V, and f_5^V are defined here in form of the coefficients of the terms $\epsilon^{\mu\alpha\beta\rho}q_{2\rho}$ and $\epsilon^{\mu\alpha\beta\rho}(q_1-q_2)_\rho$, respectively, and are described by the equation 45.

$$\begin{aligned}
h_3^\gamma(j) &= -N_f \frac{e^2 Q_j^2 g_{aj}}{2\pi^2 \sin\Theta_w \cos\Theta_w} I_3^\gamma;\\
h_3^Z(j) &= -N_f \frac{e^2 Q_j^2 g_{vj} g_{aj}}{4\pi^2 \sin^2\Theta_w \cos^2\Theta_w} I_3^Z;\\
f_5^\gamma(j) &= N_f \frac{e^2 Q_j^2 g_{vj} g_{aj}}{4\pi^2 \sin^2\Theta_w \cos^2\Theta_w} I_5^\gamma.
\end{aligned} \tag{45}$$

The expressions for I_3^γ, I_3^Z, I_5^γ in equation (45) can be taken from [26]; $g_{aj} \equiv I_{3,j}$ is the third isospin projection of considering j-th particle and $g_{vj} \equiv I_{3,j}(1-4|Q_j|\sin\Theta_w)$.

In order to calculate the loop amplitudes of the $\gamma \to Z\gamma$, $Z \to Z\gamma$ and $\gamma \to ZZ$ processes, the following

Lagrangian is used:

$$L = -eQ_j A^\mu \overline{f}_j \gamma_\mu f_j - \frac{e}{2c_w s_w} Z^\mu \overline{f}_j (\gamma_\mu g_{vj} - \gamma_\mu \gamma_5 g_{aj}) f_j - \\ - \frac{e}{2c_w s_w} Z^\mu \overline{f}_1 (\gamma_\mu g_{v12} - \gamma_\mu \gamma_5 g_{a12}) f_2. \tag{46}$$

The effective couplings h_3^V, and f_5^V are calculated as the coefficients of the terms $\epsilon^{\mu\alpha\beta\rho} q_{2\rho}$ and $\epsilon^{\mu\alpha\beta\rho}(q_1 - q_2)_\rho$ in the amplitude, respectively. The calculation is performed using the Passarino and Veltman reduction procedure [28]. The idea is to reduce the one-loop integrals to a linear combination of the Passarino and Veltmann functions. Due to the additivity of the amplitude, the sum over all the particles participating in the process is calculated in order to take into account the corresponding contributions:

$$h_3^V(\text{Total}) = \sum_{j \in \text{Particles}} h_3^V(j); \qquad f_5^V(\text{Total}) = \sum_{j \in \text{Particles}} f_5^V(j). \tag{47}$$

Each of $h_3^\gamma(j)$, $h_3^Z(j)$ and $f_5^\gamma(j)$ couplings depends on the masses of the particles arising in the loop (Figure **??**). The SM contributions to this couplings arises only from the three families of leptons and quarks, whereas in the MSSM the chargino and neutralino also contribute at the one-loop level. In this context, the parameter M is a free parameter of the MSSM model (characteristic energy scale of the BSM particles) and the masses of Chargino can be expressed as follows [29]:

$$M^2_{\chi 1,2} = \frac{1}{2}\Big(M^2 + \mu^2 + 2m_W^2 \mp \sqrt{\big(M^2 + \mu^2 + 2m_W^2\big)^2 - 4\big(M\mu - m_W^2 \sin(2\beta)\big)^2}\Big). \tag{48}$$

Obtained results for SM and MSSM models are shown in Figure 17, where the (a,b,c) plots illustrate the values of the couplings h_3^γ, h_3^Z and f_5^γ respectively as functions of $\sqrt{s}$. The predicted absolute values of h_3^γ, h_3^Z and f_5^γ couplings for SM and MSSM (M=150 GeV) models are presented in Tables 2 and 3 respectively.

Table 2: Absolute values of the effective couplings for the SM.

$\sqrt{s}$, TeV	$h_3^\gamma \cdot 10^8$	$h_3^Z \cdot 10^8$	$f_5^\gamma \cdot 10^8$
7	2.59	0.95	0.95
8	1.62	0.59	0.59
13	0.29	0.10	0.11
14	0.22	0.08	0.08

Table 3: Absolute values of the effective couplings for the SM + Chargino. **The tables are too big everywhere**

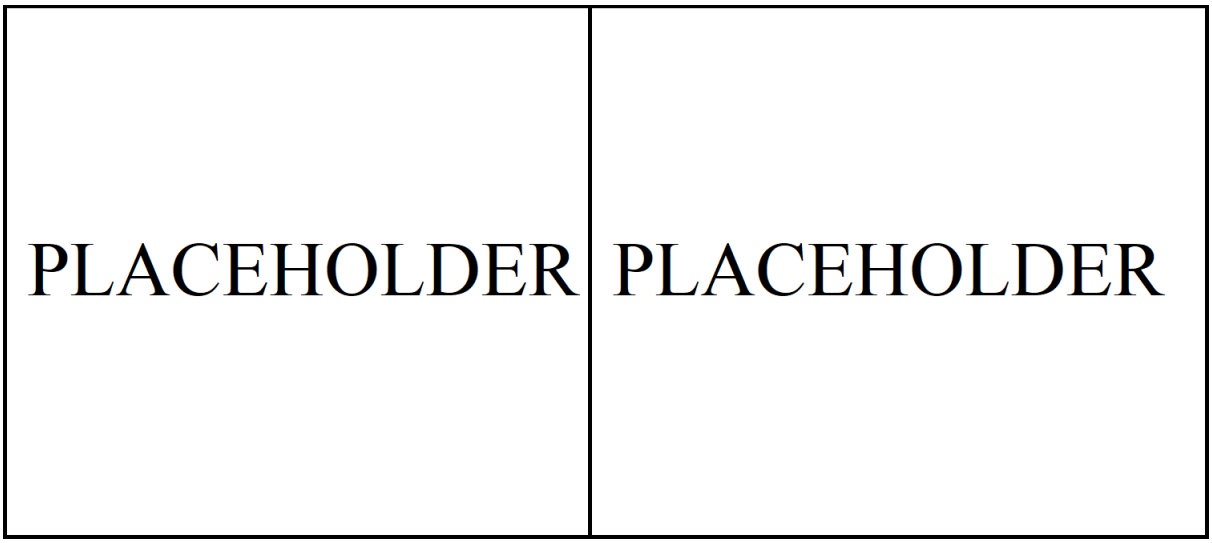
$\sqrt{s}$, TeV	$h_3^\gamma \cdot 10^8$	$h_3^Z \cdot 10^8$	$f_5^\gamma \cdot 10^8$
7	2.16	9.53	0.90
8	1.34	7.16	0.62
13	0.23	2.58	0.18
14	0.18	2.22	0.14

The allowed regions of the effective couplings h_3^γ, h_3^Z and f_5^γ, extracted from the experiments at the moment, are shown in table 4.

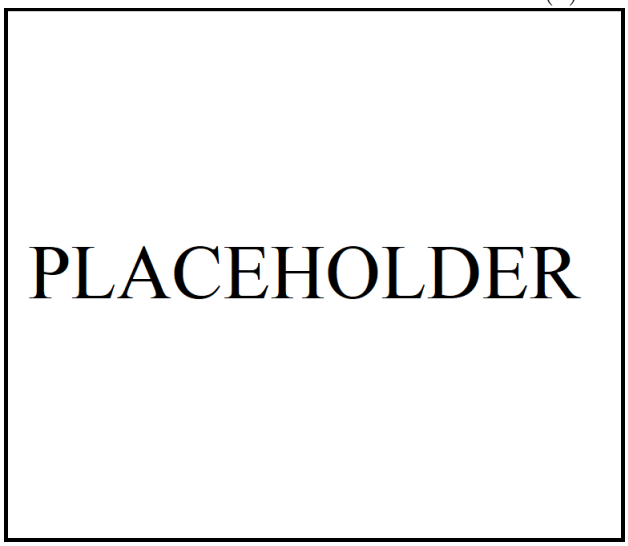
Thus the experimental sensitivity at the moment is not enough to exclude this model with $M = 150$ GeV. **So we have two failed cases now - is that reasonable?**

D Old text

To obtain constraints on the f_4^Z coupling, **we** perform numerical calculations using the `LoopTools` package for Wolfram Mathematica [30]. The value of f_4^Z is computed as a function of the 2HDM parameters introduced earlier, namely the masses of the heavy neutral and charged Higgs bosons (m_2, m_3, $m_{H^\pm}$) and the scalar



(a) (b)



(c)

Figure 17: The SM and MSSM contributions to the h_3^γ (a), h_3^Z (b) and f_5^γ (c) couplings as functions of $\sqrt{s}$. **All the figures should be redone**

Table 4: Experimental constrains on the effective couplings h_3^γ, h_3^Z and f_5^γ [8, 12].

$\sqrt{s}$, TeV	h_3^γ	h_3^Z	f_5^γ
7	$[-2.9 \div 2.9] \cdot 10^{-3}$	$[-2.7 \div 2.7] \cdot 10^{-3}$	$[-1.6 \div 1.5] \cdot 10^{-2}$
8	$[-9.5 \div 9.9] \cdot 10^{-4}$	$[-7.8 \div 8.6] \cdot 10^{-4}$	$[-3.8 \div 3.8] \cdot 10^{-3}$
13	$[-3.7 \div 3.7] \cdot 10^{-4}$	$[-3.2 \div 3.3] \cdot 10^{-4}$	$[-6.8 \div 7.5] \cdot 10^{-4}$

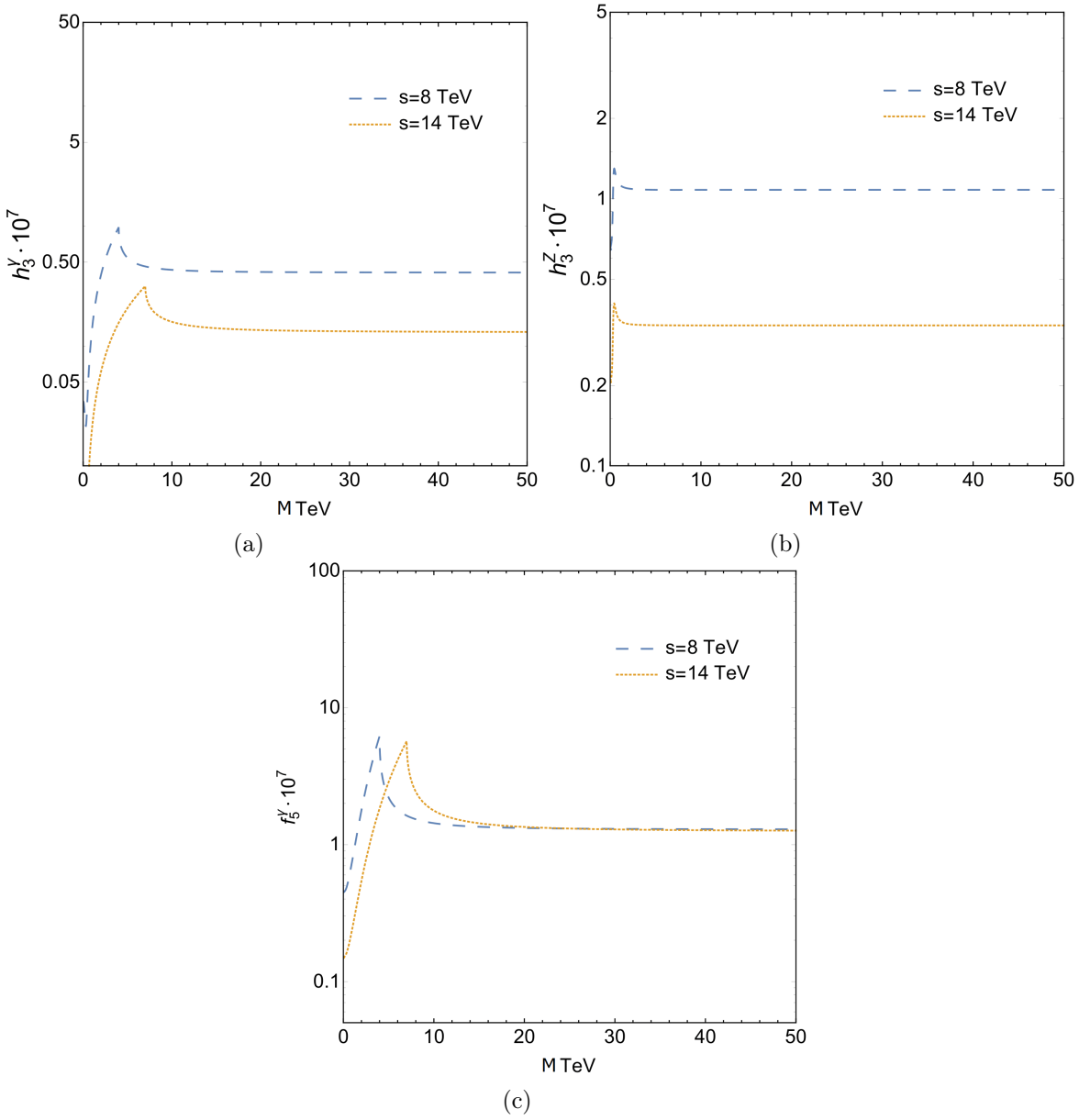


Figure 18: The SM and MSSM contributions to the h_3^γ (a), h_3^Z (b) and f_5^γ (c) couplings as functions of M.

mixing angles (α_2, α_3). Additionally, the coupling depends on the squared momentum transfer q^2, which is not a parameter of the model but a kinematic variable. A detailed discussion of how this dependence is treated in the context of experimental comparison is given in Section 3.

The dependence of the f_4^Z coupling on the scalar mixing angles enters through a multiplicative prefactor $(T_{11}T_{12}T_{13})$. This angular factor is given analytically as $\sin^2(\alpha_2)\cos(\alpha_2)\sin(\alpha_3)\cos(\alpha_3)$, and does not depend on α_1. Its role is to rescale the amplitude of the loop-induced f_4^Z vertex, while leaving the functional dependence on the masses and momentum transfer unchanged.

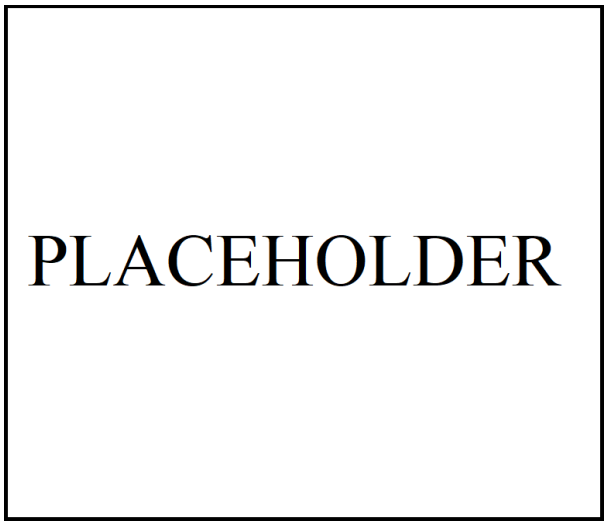
Figure 19a shows the two-dimensional distribution of the angular prefactor as a function of α_2 and α_3. From this plot, one can identify the region of maximum coupling strength ($\alpha_2 \approx 0.95$ and $\alpha_3 \approx 0.78$). To obtain the strongest possible constraints on the model parameters, **we** fix the angular values to those that maximize the prefactor throughout the analysis.

Let us assume, for simplicity, that the mass of the third Higgs boson (m_3) is related to the second via the electroweak vacuum expectation value, such that $m_3 = \sqrt{m_2^2 + v^2}$. As a result, the f_4^Z coupling effectively depends only on two free parameters: the mass of the second neutral Higgs boson m_2 and the squared momentum transfer q^2. Figure 19b presents the two-dimensional distribution of the absolute value of the coupling in the (m_2, q) parameter space. The plot indicates that f_4^Z becomes significant in the region $0 < m_2 < 2$ TeV and $0 < q < 1$ TeV. The graph also includes projected exclusion regions based on current and expected experimental sensitivities at two integrated luminosities.

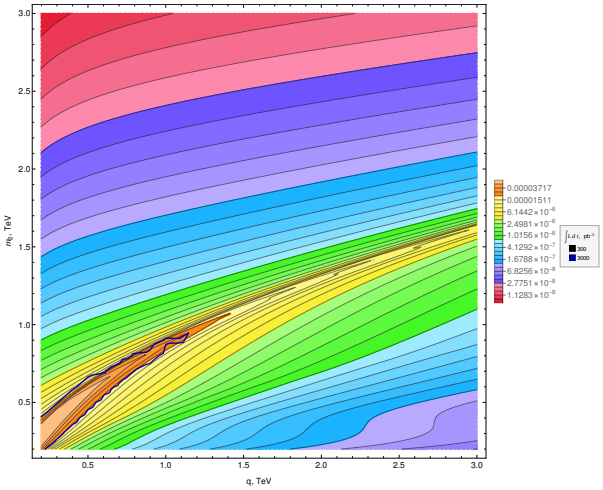
To further illustrate the dependence of the f_4^Z coupling on individual parameters, one-dimensional projections were constructed: $f_4^Z(m_2)$ at fixed values of q (Figure 19c), and $f_4^Z(q)$ at fixed m_2 (Figure 19d). These plots include expected exclusion limits from current and future LHC luminosities: the red and blue shaded regions correspond to integrated luminosities of 300 and 3000 fb^{-1}. Colored lines represent different fixed values of the second parameter.

Figures 19e and 19f show the q-dependence of f_4^Z for two benchmark values of the mixing angle α_2: the optimal value $\alpha_2 = \alpha_{\text{opt}}$ and a representative off-optimum value $\alpha_2 = 0.5$. These plots illustrate the rescaling role of the mixing angular parameters.

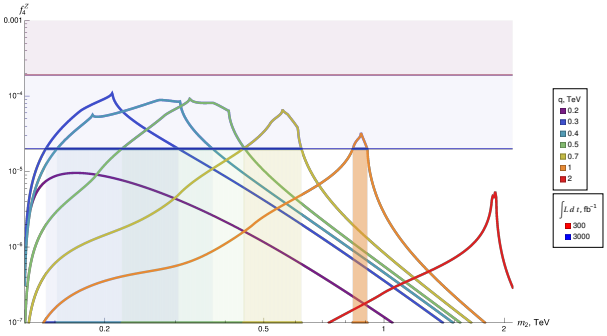
In the next section, **we** address the dependence of f_4^Z on q^2 and present a model-independent approach to remove it. This allows for direct comparison with experimental limits and for deriving robust constraints on the 2HDM parameter space.



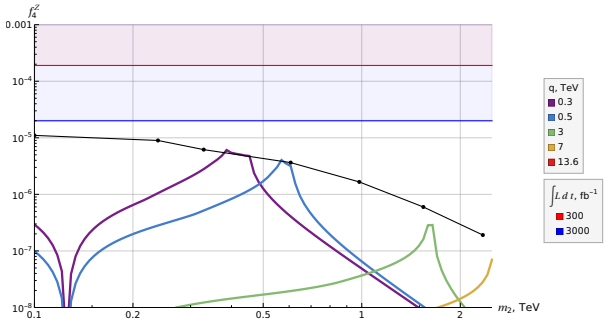
(a) Angular prefactor: $|f_4^Z|$ vs. α_2, α_3



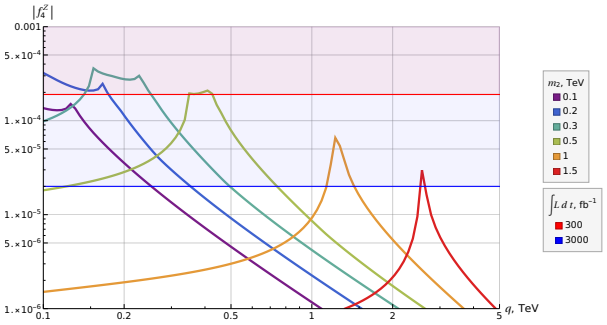
(b) $|f_4^Z|$ vs. m_2, q



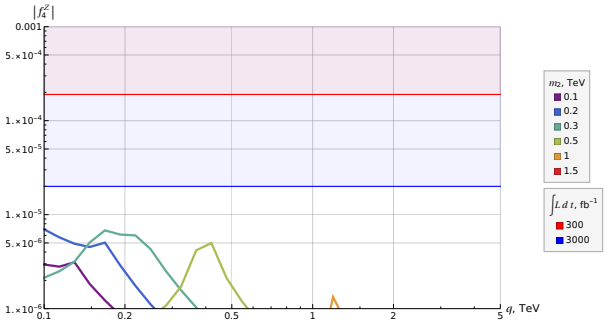
(c) $|f_4^Z|$ vs. m_2 for various q



(d) $|f_4^Z|$ vs. q for various m_2



(e) $|f_4^Z(q)|$ at $\alpha_2 = \alpha_{\text{opt}}$



(f) $|f_4^Z(q)|$ at $\alpha_2 = 0.5$

Figure 19: Overview of the f_4^Z coupling behavior in the 2HDM. Panel (a) shows the angular prefactor; (b) the dependence on m_2 and q; (c–d) one-dimensional slices over m_2 and q with fixed parameters; (e–f) q-dependence at different mixing angles. Shaded regions correspond to expected experimental sensitivity at 300 and 3000 fb^{-1}.